🦁 SmartZoo AI - Complete Project Documentation for Social Media

📋 Project Overview

SmartZoo AI is a bilingual (English/Urdu) animal classification system that identifies 90+ animal species from images with 85% accuracy. It combines:

- Computer Vision (MobileNetV2) for image recognition
- Large Language Model (Groq Llama 3.3 70B) for AI chat
- RAG Knowledge Base for factual animal information
- Voice Input with Whisper speech-to-text
- Bilingual Support with Helsinki-NLP translation

🔧 How I Built It - Technical Details

Phase 1: Data Collection & Preparation

- Dataset: 90 Animal Species Dataset from Kaggle (10,000+ images)
- Data Split: 80% training (8,000 images) | 20% validation (2,000 images)
- Image Preprocessing: Resized to 224x224 pixels, normalized to [0,1]

Phase 2: Model Training (MobileNetV2)

- Architecture: MobileNetV2 pre-trained on ImageNet (transfer learning)
- Framework: TensorFlow 2.x / Keras
- Training Strategy: 2-phase fine-tuning
    - Phase 1: Trained classifier head only (10 epochs, LR: 0.001)
    - Phase 2: Unfroze last 50 layers, fine-tuned (15 epochs, LR: 0.0001)
- Regularization: Dropout (0.3-0.5), L2 regularization (1e-4)
- Hardware: Kaggle Tesla T4 GPU (~2 hours training)
- Results: 85% validation accuracy, 92% training accuracy

Phase 3: RAG Knowledge Base System

- Data Source: JSON file with 70+ animal entries (habitat, diet, conservation, fun facts)
- Retrieval: Keyword + tag-based matching to detect animal from user query
- Augmentation: Retrieved fact injected into LLM prompt
- Generation: Groq Llama 3.3 70B generates conversational response

Phase 4: Web Application (Streamlit)

- Image Upload: Real-time classification with confidence chart
- AI Chat Assistant: Conversational Q&A about detected animals
- Voice Input: Groq Whisper for speech-to-text
- Translation: Helsinki-NLP (opus-mt-en-ur) for English→Urdu
- RAG Search: Knowledge base query before image upload
- Dark Theme: Custom CSS with emerald green & midnight gold

Phase 5: Deployment

- Platform: Hugging Face Spaces (Free tier)
- SDK: Streamlit
- Files: app.py, requirements.txt, animals_data.json, model files

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📊 Results Summary

Metric	Score
Validation Accuracy	85%
Training Accuracy	92%
Inference Time	<0.1 seconds
Animal Classes	90+
Languages Supported	English, Urdu

RAG Knowledge Base	70+ animals